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Top Skills

Hybrid Emulation
Generative AI
OOAD

Languages

Hindi (Native or Bilingual)
English (Professional Working)

Honors-Awards

2nd Rank in Bihar Combined
Competitive Exam
School Topper in Xth CBSE Exam

Publications

Innovative flow for reuse of IP C
models from IVAHD2.0 Design to
Virtual Platform Simulator for pre-
silicon software development
Accelerating SystemC SoC
Development Using Text Based
Netlist Connectivity Specification
Enabling Reset in SystemC based
Virtual Platforms

Patents

Method and system for emulating an
image processing system

Sanjay Kumar Singh

Senior Software Architect | SoC Virtual Platforms & Hardware-
Software Systems | Exploring Gen AI for EDA | NIT Trichy | BITS
Pilani

Bengaluru, Karnataka, India

Summary

I architect complex hardware–software hybrid systems at the intersection of virtual platforms, silicon bring-up, and enterprise engineering. With 22+ years of experience, I build infrastructure that enables semiconductor teams to move faster, scale smarter, and reduce time-to-silicon. As a Senior Architect, I lead a team of engineers and partner closely with customer, partners and stakeholders to define and deliver simulation and emulation platforms that compress hardware development cycles. My work spans the full stack—from IP modeling and SystemC/TLM 2.0 methodologies to hybrid emulation infrastructure, virtual platform integration, and production release to end customers. Over the years, I've built systems many believed couldn't be simulated: DDR4/DDR5 and LPDDR4/LPDDR5 memory transactors, CSI/DSI virtual device models, and full SoC virtual platforms used across the chip development lifecycle by global engineering teams. More recently, I've been investing deeply in Generative AI—exploring how LLM-powered tooling and AI-assisted workflows can transform how engineers design, verify, and debug complex systems. I believe the next wave of EDA and virtual platform innovation will be shaped by AI, and I am actively building in this space. I hold a B.Tech in Computer Science from NIT Trichy and an M.S. in Microelectronics from BITS Pilani.

Experience

Cadence Design Systems

12 years 3 months

Senior Software Architect

July 2024 - Present (1 year 11 months)

Bengaluru

- Deployed hybrid emulation solutions at tier-1 SoC customers, enabling full OS boot (Linux and Android) on virtual and hybrid platforms.

- Owned full-stack delivery across virtual IP modeling, HW/SW transactor development, hybrid platform integration, and embedded software bring-up, providing end-to-end ownership from pre-silicon software development to customer deployment.
- Currently pioneering a multi-threaded simulation framework addressing one of SystemC's most longstanding architectural limitations, with potential to significantly improve simulation throughput for complex SoC designs.

Software Architect

October 2020 - July 2024 (3 years 10 months)

Bengaluru

Led end-to-end delivery of the Helium Virtual and Hybrid Platform Catalog, built ground-up with a team of engineers. The platform spans UFS, NVMe, Ethernet, and USB IP models; hybrid transactors for AMBA buses; and a complete embedded software stack—including bare-metal, Linux, and Android bring-up.

Enabled pre-silicon software development for enterprise customers across mobile and hyperscaler SoC domains, compressing time-to-market by allowing software teams to begin development months ahead of first silicon availability. Designed and implemented a virtual-to-RTL gear-shift execution platform, enabling workloads to start execution on virtual CPUs and seamlessly transfer architectural state to RTL CPUs—allowing customers to run real, silicon-representative benchmark tests on hybrid platforms.

Senior Principal Software Engineer

July 2016 - October 2020 (4 years 4 months)

Bengaluru

Architected and developed Cadence Smart Memory, a high-performance coherent memory solution that significantly accelerated hybrid RTL + SystemC platform execution, removing a critical bottleneck in mixed-signal and mixed-abstraction simulation environments.

Built a comprehensive virtual device model library covering CSI-2 (Camera), DSI (Display), HDMI, and I2C protocols, enabling realistic pre-silicon system validation across imaging and display subsystems.

Principal Solutions Engineer

March 2014 - July 2016 (2 years 5 months)

Bengaluru

Led hybrid platform integration and full-stack software bring-up for multiple in-house and customer ARMv8-based mobile SoCs, delivering system enablement from bare-metal through OS-level bring-up.

Texas Instruments

8 years 6 months

Technical Lead

February 2012 - March 2014 (2 years 2 months)

Bangalore

Led end-to-end execution of the IVAHD 2.0 Virtual Platform at Texas Instruments, one of the industry's pioneering SystemC adoption projects, owning the full lifecycle from requirements definition and IP team enablement through platform integration and customer release.

Drove TI's strategic transition from proprietary SSI-based modeling to OSCI SystemC, establishing a foundational architectural shift that guided all subsequent virtual platform initiatives at TI Bangalore.

Solved first-of-its-kind SystemC integration challenges, including runtime pin binding, synchronization, and reset handling—creating reusable patterns that became reference implementations for the broader virtual platform community. Developed a SystemC Reset Infrastructure using SystemC 2.3 process control APIs, and presented the solution at ISCUG 2013 (International SystemC User Group), earning industry-level recognition for the approach.

Sr Software Engineer

August 2007 - January 2012 (4 years 6 months)

Bangalore

Developed cycle-accurate models for TI Curie-class SoCs (Curie, Primus), meeting an aggressive >95% cycle-accuracy target using event-based simulation techniques.

Built detailed cycle-accurate models for the TI CBA subsystem, including SCR, M2M Bridge, and CBA (VBSUM/VBUSP) bus infrastructure.

Conceptualized and developed debug infrastructure for Data Breakpoint and UMR detection, significantly improving model debugging efficiency and code quality.

Software Engineer

October 2005 - July 2007 (1 year 10 months)

Bangalore

Joined the Simulation Team at Texas Instruments, developing a Simulation Toolkit and connection drivers based on TI's proprietary SSI modeling standard.

Built tooling to help IP C-model developers rapidly prototype models from design specifications and integrate them into TI's base simulation platforms. Analyzed diverse C-model architectures to identify appropriate integration hooks and entry points within existing simulation frameworks.

Developed code generators using scripting languages to automatically produce C++ model scaffolding from structured input descriptions, improving consistency and developer productivity.

Siemens Information Systems Ltd

Software Engineer

July 2004 - September 2005 (1 year 3 months)

Bangalore

Joined Siemens from college as part of the RTS team in the Power Sector domain. As a team member, I contributed to the development of Windows-based applications for managing large-scale customer data, with customer counts in the millions.

Fresh out of college, I quickly ramped up on C#, .NET, XML, and ASP.NET, applying these technologies to deliver production-quality software. Alongside my core work, I also explored game development using C#, strengthening my hands-on programming skills and problem-solving abilities.

Education

National Institute of Technology, Tiruchirappalli

B. Tech., Computer Science and Engineering · (2000 - 2004)

Birla Institute of Technology

MS, Microelectronics · (2009 - 2010)

Model School Dalmianagar

Xth · (1986 - 1998)